

A Counterexample to Arrhenius's Sixth Impossibility Theorem

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Abstract

Gustaf Arrhenius's sixth impossibility theorem says that no population axiology satisfies Egalitarian Dominance, General Non-Extreme Priority, Non-Elitism, Weak Non-Sadism, and the Weak Quality Addition Condition. Teruji Thomas has noted a gap in the proof, and Elliott Thornley has restated the last condition with two quantifiers reversed so that the proof goes through. I show that the gap cannot be closed: the theorem as stated is false. A complete, transitive axiology on Arrhenius's own integer welfare levels satisfies all five conditions in their exact formulations. It ranks populations first by whether their misery exceeds what their excellent lives can offset, and otherwise by total welfare; it accepts the Repugnant Conclusion and rejects the Very Repugnant Conclusion, the combination the theorem was built to rule out. The repaired theorem is true, and I show what its extra premise buys. The four remaining conditions force only a background-relative Very Repugnant Conclusion; for every axiology on which misery can be compensated by lives barely worth living, Weak Quality Addition in either form entails the Quality Addition Condition the sixth theorem was meant to weaken; and the strengthened condition excludes, by stipulation, axiologies on which the quantity of misery that marginal lives can never make up for varies with the population it is added to. Every theorem stated is verified in Lean 4, and the counterexample was found by a search method described here.

I The theorem

Population ethics has a body of impossibility theorems that do for the comparison of differently sized populations what Arrow's theorem did for voting: each lists conditions on a betterness ordering of populations, each apparently undeniable, and shows that no ordering satisfies them all. The problem they formalize is Parfit's (1984, ch. 17; 2016), that the ways of avoiding the Repugnant Conclusion each cost something no less repugnant. After Ng (1989) and Blackorby and Donaldson (1991), Gustaf Arrhenius proved six (Arrhenius 2000b, 2000a, 2003, 2009, 2011), now the standard reference point (Greaves 2017), and regards the sixth as best, because its conditions are logically the weakest. This paper shows that it is false, and that the repair which makes it true adds exactly one stipulation: that the quantity of misery which lives barely worth living can never make up for must be the same whatever population it is added to. The theorem says:

The Sixth Impossibility Theorem. There is no population axiology which satisfies the Egalitarian Dominance, the General Non-Extreme Priority, the Non-Elitism, the Weak Non-Sadism, and the Weak Quality Addition Condition. (Arrhenius 2011, 9)

The theorem has been treated as sound. The Oxford Handbook of Population Ethics takes it as a point of departure (Arrhenius et al. 2022), and Arrhenius’s own most recent statement, this year, gives the five conditions in the same form, with Weak Quality Addition beginning “For any population X, there is a perfectly equal population with very high positive welfare, and a very negative welfare level, and a number of lives at this level” (Arrhenius 2026, sec. 2 and n. 10). Its one sustained critic accepts the proof and contests a background assumption: Erik Carlson argues that the theorem depends on Finite Fine-Grainedness, that any two welfare levels are joined by finitely many slight differences, and grants that “Arrhenius’s proof shows that no population axiology satisfies the exact versions of his five adequacy conditions, given that the order of welfare levels assumed in the proof is fine-grained and A-discrete” (Carlson 2022, sec. 1). Two readers have noticed that something is wrong with the proof itself. Teruji Thomas, in an unpublished reconstruction of all six theorems, found that the published argument establishes only a weaker conclusion than the one stated, and noted in print that “the proof of his favoured sixth theorem contains an error in the derivation of the ‘Restricted Quality Addition Condition’ (Arrhenius, 2011, Lemma 1.3). As far as I know, one has to slightly strengthen the premisses of the theorem, in a way unlikely to cause further controversy” (Thomas 2018, n. 4; cf. Thomas 2016). Thornley, following Thomas, restates Weak Quality Addition with its first two quantifiers reversed, remarks that “the Sixth Impossibility Theorem actually requires the slightly stronger condition stated here,” and adds that “the stronger version remains a compelling adequacy condition” (Thornley 2021, n. 7).

Thomas’s remark implies that the theorem, not only its proof, has to change; but neither he nor Thornley shows that it does, and the strengthening has been treated as a formality. It is not. The theorem as Arrhenius states it is false, and false on the integer-indexed welfare levels that Carlson concedes to it. There is a complete, transitive, anonymous welfarist axiology on exactly those levels that satisfies all five conditions in the exact formulations Arrhenius gives them. I call it the *capacity view*. It ranks populations first by whether the misery they contain exceeds what their excellent lives can offset, and otherwise by total welfare. It accepts the Repugnant Conclusion, as total utilitarianism does, and rejects the Very Repugnant Conclusion, which is the combination of attitudes the move from Arrhenius’s earlier theorems to the sixth was designed to make untenable. So the sixth theorem does not refute the position it was built to refute.

The strengthened condition does make the theorem true; Thomas proved it in his note, and the proof is machine-checked here. What I add is an account of what the strengthening buys. In its original form Weak Quality Addition says that, for any population, there is some quantity of very bad lives such that adding them together with any number of lives barely worth living is no better than adding some number of excellent lives. The strengthened condition fixes the quantity of very bad lives first and then quantifies over populations. The difference is whether the amount of misery that marginal lives can never make up for may depend on how much excellence the population already contains. The capacity view says it may, and the four other conditions cannot force otherwise: what they entail is a Very Repugnant Conclusion relative to a background chosen after the quantity of misery is fixed, and nothing stronger. Whether the new premise is as “compelling” as the old is therefore the question

whether capacity views deserve to be ruled out by an adequacy condition rather than by argument. The counterexample was found by a search method (§5), and every theorem here is verified in the Lean 4 proof assistant (§6).

2 The framework and the five conditions

Arrhenius’s framework is this (Arrhenius 2011, 4–7). Welfare levels are quasi-ordered, with a neutral level 0; Arrhenius assumes *Discreteness*, that between any two levels there are only finitely many others, arguing that nothing turns on it since one can pass to a discrete subset of a dense order (Arrhenius 2011, 6). Given Discreteness, levels are indexed by integers, and $w + 1$ is “slightly higher” than w . Following Thomas (2016) I identify a population with its welfare distribution and write $n \cdot x$ for n lives at level x and $P + Q$ for union; Arrhenius’s own exact formulations write $A \subset W_x$, $N(A) = n$ for what I write $n \cdot x$, and his rider “other things being equal” licenses the identification.¹ An *axiology* is a reflexive, transitive, not necessarily complete relation “at least as good as”; “better than” is its strict part.

The conditions refer to a “very high” level, a “very low positive” range $R(1, y)$ (the levels from 1 to y), and a “very negative” level. In the exact formulations these are quantified over inside each condition rather than fixed in advance, which is what makes the formulations exact; I give those (Arrhenius 2011, 7–9), since the result depends on them. A range $R(x, y)$ is, by Arrhenius’s definition, a union of at least three levels (Arrhenius 2011, 7 n. 8).

Egalitarian Dominance (ED). If $N(A) = N(B)$, every member of B has welfare below x , and $A = n \cdot x$, then A is better than B .

General Non-Extreme Priority (GNEP). For any level z there are a positive level u , a positive range $R(1, y)$ with $u > y$, and a number $n > 0$ such that for any $x \geq u$, any $B \subset R(1, y)$ with $N(B) = n$, and any population E : $n \cdot x + 1 \cdot z + E$ is at least as good as $B + 1 \cdot (z + 1) + E$.

Non-Elitism (NE). For any levels x, y with $x - 1 > y$ there is $n > 0$ such that for any $D \subset R(y, x)$: $(n + 1) \cdot (x - 1) + D$ is at least as good as $1 \cdot x + n \cdot y + D$.

Weak Non-Sadism (WNS). There are a level $x < 0$ and a number n such that for any level $y > 0$, any $B \subset W_y$, and any population C : $B + C$ is at least as good as $n \cdot x + C$.

Weak Quality Addition (WQA). For any population X there are a level $x < 0$, positive ranges $R(u, v)$ and $R(1, y)$ with $u > y$, and numbers $n, m > 0$ such that for any $z \geq u$ and any $B \subset R(1, y)$: $n \cdot z + X$ is at least as good as $B + m \cdot x + X$.

Informally: ED says a perfectly equal population is better than any same-sized population every member of which is worse off. GNEP says that lowering one person’s welfare by a step is never worse than

¹Every condition is invariant under replacing lives by others at the same level, so nothing is lost by working with distributions.

failing to raise a fixed number of others from very low positive to very high welfare, whatever else is true of the population. NE says that one person's small loss can be compensated by enough others' gains to the level she falls to, when the rest of the population lies between those levels. WNS says some number of very bad lives is no better an addition than any number of lives worth living. WQA says that for any population there is some quantity of very bad lives such that adding them, with any number of lives barely worth living, is no better than adding some number of very good lives.

Three features of WQA matter. First, the comparison is of additions to a *background* X , and the witnesses are chosen after X is given. Second, B ranges over low-range populations of every size, so the m bad lives cannot be compensated by any number of marginal lives. Third, WQA weakens Arrhenius's Quality Addition Condition, which drops the m bad lives and says that for any X some number of very good lives added to X is at least as good as any number of marginal lives added to X . Quality Addition is the denial of a background-relative Repugnant Conclusion, which total utilitarians accept; Weak Quality Addition is the denial of a background-relative Very Repugnant Conclusion, that for any population of excellent lives there is a better one consisting of some very bad lives together with enough lives barely worth living. Arrhenius weakened the condition because avoiding the Very Repugnant Conclusion is, as he and his coauthors later put it, "intuitively more compelling" than avoiding the Repugnant Conclusion (Arrhenius et al. 2021, 119): the sixth theorem is meant to bind even those who, like Tännsjö (2002), Huemer (2008) and a growing group (Zuber et al. 2021; Spears and Budolfson 2021), have made their peace with the Repugnant Conclusion.

3 The capacity view

Fix a "very high" level $A \geq 4$, so that the very low positive levels 1, 2, 3 lie below it. For a population P let $H(P)$ be the number of lives at or above A (the *excellent* lives), $T^-(P)$ the total welfare of the lives with negative welfare, and $T(P)$ the total welfare. Let $I(P) = H(P) + T^-(P)$, and for a *baseline capacity* $c \geq 0$ let $J(P) = \min(0, I(P) + c)$. I measures misery against excellence, each excellent life able to offset one unit of negative welfare; c is misery that any population, the empty one included, can absorb; J is zero when excellence and baseline cover the misery and otherwise equals the uncovered misery. The *capacity view* ranks populations lexically by (J, T) :

P is at least as good as Q if and only if either $J(P) > J(Q)$, or $J(P) = J(Q)$ and $T(P) \geq T(Q)$.

This is complete and transitive, being the lexicographic order on pairs of integers, and it is welfarist and anonymous. A population whose misery exceeds its capacity to redeem it is worse than any whose misery does not, and among the former the smaller excess is better; among the latter, total welfare decides. The exchange rate of one excellent life per unit of misery is a convenience; any positive weight on H works. The simplest member of the family has $c = 0$, and the reader may take that as the official countermodel.

The view is lexical, hence discontinuous: a unit of uncovered misery outweighs any quantity of total welfare. With $c = 0$, a billion lives at level 3 together with one life at -1 has $J = -1$ and is worse

than the empty population; with $c \geq 1$ the baseline absorbs the unit and the consequence disappears (§4). The view implies the Repugnant Conclusion, since misery-free populations are ranked by total welfare. It rejects the Very Repugnant Conclusion: for $m|z| > c$, the population $m \cdot z + N \cdot 3$ has $I + c < 0$ and is worse than $n \cdot A$, which has $J = 0$, for every N .

It has a close relative in the literature. In the paper that notes the gap, Thomas sketches a “toy model” with “a sufficientarian flavour”: integer levels, a sufficiency level S , and populations “ranked, in the first instance, by the number of very good lives minus the number of very bad lives,” with ties broken by total welfare (Thomas 2018, sec. 4; cf. Qizilbash 2005; Knapp 2007; Parfit 2016). He offers it for another purpose: it rejects the Repugnant Conclusion and violates Non-Elitism, but only at the threshold S , so that if S is vague the violations are borderline. The capacity view is Thomas’s model with two changes: misery is weighted by welfare rather than counted, and, decisively, the first layer is capped at zero. The cap is what turns a determinate violation of Non-Elitism into none, since the one transfer that lowers I is licensed only in misery-free populations, where the capped layer is already zero; and the cap is what makes the view accept the Repugnant Conclusion.

Theorem 1. For every $A \geq 4$ and $c \geq 0$, the capacity view satisfies ED, GNEP, NE, WNS and WQA in the exact formulations of §2.²

Each of the first four conditions compares two populations that differ by a *transfer*: lives are moved between levels or added. Write ΔI and ΔT for the change in I and T from the population the condition says is no better to the one it says is at least as good. If $\Delta I \geq 0$ and $\Delta T \geq 0$ the view agrees with the condition, since J is monotone in I and c cancels; backgrounds drop out, because I and T are additive over union.

Egalitarian Dominance. $A = n \cdot x$, every member of B is below x , $N(B) = n$. Then $T(A) = nx > T(B)$. If $x \geq A$ then $H(A) = n \geq H(B)$; if $x < A$ then $H(B) = 0 = H(A)$. And $T^-(A) \geq T^-(B)$: if $x < 0$ every member of B is negative and below x , so $T^-(B) = T(B) < nx = T^-(A)$; if $x \geq 0$ then $T^-(A) = 0 \geq T^-(B)$.

General Non-Extreme Priority. Given z , choose $u = A$, $y = 3$, $n = 2$. Two excellent lives raise H by 2; the person moved from $z + 1$ to z lowers H by 1 if $z + 1 = A$ and lowers T^- by 1 if $z + 1 \leq 0$; B contributes nothing to I . So $\Delta I \geq 0$, and $\Delta T \geq 2A - 7 > 0$.

Non-Elitism. Given $x - 1 > y$, choose $n = 1$: compare $1 \cdot x + 1 \cdot y + D$ with $2 \cdot (x - 1) + D$, $D \subset R(y, x)$. $\Delta T = x - 2 - y \geq 0$. If $x - 1 \geq A$ the two lives at $x - 1$ are excellent, $\Delta H \geq 0$, and $\Delta T^- \geq 0$ since only the life at y can be negative and it is raised. If $x < A$ then $\Delta H = 0$ and $\Delta T^- \geq 0$; the only case needing care is $x = 0$, where two lives at -1 replace one at 0 and one at $y \leq -2$, and $\Delta T^- = |y| - 2 \geq 0$. The remaining case is $x = A$, where an excellent life is lost: $\Delta H = -1$. If $y < 0$ the life raised from y gives $\Delta T^- = |y| \geq 1$, so $\Delta I \geq 0$. If $y \geq 0$ then $D \subset R(y, A)$ contains

²Arrhenius’s 2026 statement gives the informal conditions with “better than” in place of “at least as good as,” noting that the exact versions are the weaker ones (Arrhenius 2026, n. 10). The capacity view satisfies the strict forms too: ED and WQA are strict below, GNEP and WNS have $\Delta T > 0$, and NE is strict once its witness is raised from 1 to 2. The machine-checked statements are the non-strict ones.

no negative lives, both populations have $J = 0$, and T decides, and T does not fall. This is the one place where I can fall under a licensed transfer, and it falls only in a population whose J is already zero and stays zero. The restriction on D makes this work; Arrhenius’s General Non-Elitism, used in his fifth theorem, drops it, and the capacity view violates General Non-Elitism, as it must, since the fifth theorem is valid.

Weak Non-Sadism. Choose $x = -1$, $n = 1$: compare $1 \cdot (-1) + C$ with $B + C$, B at some $y > 0$. $\Delta I = H(B) + 1 \geq 1$ and $\Delta T = T(B) + 1 \geq 1$.

Weak Quality Addition. Given X , choose $x = -1$, $u = A$, $y = 3$, $n = 1$, and $m = \max(0, I(X) + c) + 1$. For any $z \geq A$ and $B \subset R(1, 3)$: $I(1 \cdot z + X) = 1 + I(X)$, while $I(B + m \cdot (-1) + X) + c = I(X) + c - m < 0$, since B contributes nothing to I . So $J(B + m \cdot (-1) + X) < J(1 \cdot z + X)$, and $1 \cdot z + X$ is better for every B , whatever its size.

The last argument displays the mechanism. WQA lets the quantity of misery depend on the background, and the capacity view uses exactly that freedom: it chooses enough misery to exceed the background’s capacity, after which no quantity of marginal lives helps, because marginal lives contribute to T and not to I . The other four conditions never require comparing a population that has exceeded its capacity favourably with one that has not, because none of the transfers they license lowers I except in populations with no misery.

So the Sixth Impossibility Theorem is false, and Arrhenius’s argument fails at exactly the step Thomas identified. In the proof of Lemma 1.3, step (12) applies WQA to the background $A_1 \cup X$ using witnesses the condition supplies for the background X (Arrhenius 2011, 18–19), where A_1 consists of excellent lives whose number depends on those very witnesses. On the capacity view, adding A_1 raises the background’s capacity, and the misery that exceeded the capacity of X no longer exceeds that of $A_1 \cup X$, so in the enlarged background the marginal lives win.

4 The repair and what it costs

In Thomas’s notation (2016, 11), the conclusion Arrhenius intended is

Very Repugnant Addition (VRA). For some population P , any level $z < 0$, and any $m, n \geq 1$, there is N such that $P + m \cdot z + N \cdot 3$ is better than $P + n \cdot A$,

and the conclusion the proof delivers is

Very Repugnant Addition* (VRA*). For any level $z < 0$ and any $m \geq 1$, there is a population P such that for any $n \geq 1$ there is N with $P + m \cdot z + N \cdot 3$ better than $P + n \cdot A$.

In VRA* the background may depend on z and m . Thomas writes that VRA* “is slightly weaker than VRA, because P is allowed to depend [on] z and m . I think that VRA* is not much less ‘repugnant’ than VRA, but then again it is not easy to understand the difference between them” (Thomas 2016, 11). Thornley’s condition, which in Thomas’s fixed-threshold setting entails the denial of VRA*, is:

Weak Quality Addition* (WQA*). There are a level $x < 0$ and a number $m > 0$ such that for any population X there are a positive level u , a range $R(1, y)$ with $u > y$, and $n > 0$ such that for any $z \geq u$ and any $B \subset R(1, y)$: $n \cdot z + X$ is at least as good as $B + m \cdot x + X$.

Thomas's corrected theorem is that ED, GNEP, NE and WNS entail VRA^* , so that the five conditions with WQA^* are jointly unsatisfiable. I have verified his proof and proved the unsatisfiability directly for Arrhenius's exact formulations as well (§6). The capacity view violates WQA^* : given x and m , take X to be $m|x|$ lives at level A , so the misery is exactly covered and both sides have $J = 0$; total welfare decides, and enough lives at level 1 in B make the right-hand side better.

Whether VRA^* is much less repugnant than VRA is the question whether WQA^* is as compelling as WQA , and the capacity view says what is at stake. The two conditions agree on every axiology satisfying:

Compensation. For any population Y , any level $z < 0$ and any m , there is K such that $Y + m \cdot z + K \cdot 3$ is at least as good as Y .

Compensation says that any finite misery added to any population can be made up for by enough lives barely worth living. Total utilitarianism satisfies it, as do critical-level views with critical level below 3 (Blackorby et al. 2005; Broome 2004), prioritarian views, and any view Archimedean in the trade between misery and marginal lives. For such views both WQA and WQA^* entail Quality Addition:

Proposition 2. If an axiology satisfies Compensation and WQA , it satisfies Quality Addition in Arrhenius's exact form: for any X there are a positive level u , a range $R(1, y)$ with $u > y$, and $n > 0$ such that $n \cdot z + X$ is at least as good as $B + X$ for every $z \geq u$ and every $B \subset R(1, y)$.

Proof. Given X , let x, u, y, n, m be the witnesses WQA supplies, and take $z \geq u$ and $B \subset R(1, y)$. Compensation applied to $X + B$ gives K with $X + B + m \cdot x + K \cdot 3$ at least as good as $X + B$. Since $y \geq 3$, $B + K \cdot 3$ lies in $R(1, y)$, so WQA gives $n \cdot z + X$ at least as good as $B + K \cdot 3 + m \cdot x + X$. Transitivity finishes. $\square$

The same argument works for WQA^* , and the converse holds given Weak Non-Sadism with a positive number of lives.³ So for a Compensation-satisfying axiology the weakening of Quality Addition to Weak Quality Addition, which distinguishes the sixth theorem from the fourth, buys nothing. With Thomas's fourth theorem, that ED, GNEP, NE and WNS entail Repugnant Addition (for any X and n there is N with $N \cdot 3 + X$ better than $n \cdot A + X$, which entails the denial of Quality Addition), this gives, in Thomas's fixed-threshold setting:

Corollary 3. ED, GNEP, NE, WNS, WQA and Compensation are jointly unsatisfiable. Given the four conditions the sixth theorem shares with the fourth, Weak Quality Addition entails that some

³Given X , let QA supply u, y, n and WNS supply x' and $n' > 0$. For $B \subset R(1, y)$, WNS with one added life at level 1 and background $B + X$ gives $B + 1 \cdot 1 + X$ at least as good as $B + n' \cdot x' + X$, and QA , since $B + 1 \cdot 1 \subset R(1, y)$, gives $n \cdot z + X$ at least as good as that; so WQA holds with $x = x', m = n'$, and since these do not depend on X , so does WQA^* .

finite quantity of misery cannot be made up for by any number of lives barely worth living.

The weakening therefore has content only for axiologies that violate Compensation, and, given the other four conditions, *commits* one to violating it. Those are the axiologies at which the sixth theorem is aimed: lexical views on which the badness of a very bad life is of a different order from the goodness of a marginal one, such as the Lexical Totalism of Carlson (2022), Thomas (2018) and Thornley (2021; cf. Thornley 2022), and the capacity view. Among them WQA and WQA* come apart, and the capacity view is the shape of the gap. WQA* says the quantity of misery that marginal lives cannot make up for can be fixed once, independently of the population it is added to; WQA says only that for each population there is such a quantity. The capacity view takes the second option, because it treats misery as redeemable by excellent lives and irredeemable by marginal ones: in a population rich in excellent lives any fixed quantity of misery is covered, and once covered it is subject to the ordinary totalist trade against marginal lives; in a population poor in excellent lives the same quantity is uncovered, and no quantity of marginal lives touches it.

Is that a structure an adequacy condition should rule out? I do not think the question has an obvious answer, and I want to resist two quick ones.

The first is that the capacity view is obviously absurd, so WQA* is as harmless as Thornley says. The view has features nobody would defend as stated: an arbitrary exchange rate, a sharp threshold, and an origin in the search for a model rather than for the truth. With $c = 0$ it also says that a billion marginal lives with one slightly bad life is worse than nothing. That consequence is worth separating from the others, because it is not a commitment of the structure. The baseline removes it: with $c \geq 1$ the empty population absorbs a unit of misery, every condition still holds, and the Very Repugnant Conclusion still fails for misery exceeding the baseline. What no member of the family can remove is uncompensable misery at the empty background once the misery exceeds c , and that is not peculiar to the capacity view. By Corollary 3, anyone who accepts the five conditions denies Compensation; Lexical Totalism in Carlson's and Thornley's versions denies it at every background, since a very negative life has a negative superior component that no number of barely-positive lives, which add only inferior goods, can restore: "adding a combination of very negative welfare lives and barely positive welfare lives means adding (r, s) with $r < 0$ " (Thornley 2021, sec. 4; cf. Carlson 2022, secs. 4–5). What the baseline shows is that the five conditions do not force the uncompensability of a small quantity of misery. They do not force the uncompensability of mild misery in any quantity either, though every member of the family as defined has it, since misery is weighted by welfare from level -1 down: a variant that counts only lives below a very negative level V , with weight $\ell - V$, satisfies the same conditions (checked numerically, not machine-checked) and makes any number of lives at -1 compensable at every background. What the five conditions force is only what Corollary 3 says, that some finite misery is uncompensable by marginal lives. The sharp threshold is likewise an artefact of taking the simplest potential: a graded version, in which a life's redemptive weight rises linearly from 0 at level 3 to 1 at level A , also satisfies the five conditions (checked numerically for $A \leq 10$; the machine-checked proof is for the step version). On that version every life above the barely-worth-living range helps redeem misery, the more the higher it lies up to A , and the structural idea is not absurd. It is

close to the thought that the Very Repugnant Conclusion is worse than the Repugnant one: someone who accepts that enough marginal lives outweigh excellent lives, but denies that they do when misery has been thrown in, is treating misery as a dimension that marginal lives do not reach. The capacity view is the simplest way of making that attitude consistent with the four other conditions, and that it must let the uncompensable quantity depend on the background is a discovery about the attitude, not an artefact of the construction.

The second quick answer is that VRA^* is as repugnant as VRA , so the weakening does not matter. Here the capacity view gives a reason to disagree with Thomas's first instinct, though I should be clear whose reason it is. VRA^* says that for any quantity of misery there is a background against which adding it together with enough marginal lives beats adding excellent lives. On the capacity view the background that does the work is rich enough in excellent lives to cover the misery, and what VRA^* then asserts is that, in a population whose excellent lives already redeem its misery, enough additional marginal lives beat additional excellent lives. That is the Repugnant Conclusion again, for a population with covered misery in it, and someone who accepts the Repugnant Conclusion has no new reason to find it repugnant. That is how the capacity view describes its own verdict. A critic of the Very Repugnant Conclusion who is not a capacity theorist will describe the same verdict differently: in some sufficiently good world, adding lives of torment with enough lives barely worth living is better than adding excellent lives, and the goodness of the surrounding world does not make the torment less. I do not think the capacity view answers that critic without presupposing itself. What it has is a diagnosis. The critic's complaint is against VRA^* , which the four conditions entail; so the critic must reject one of ED, GNEP, NE and WNS, and Weak Quality Addition in any form will not help, because the complaint is about a conclusion the four conditions reach without it. The repaired theorem tells the critic of the Very Repugnant Conclusion that the fight is elsewhere. The unrepaired theorem told the critic, wrongly, that there was no consistent position to fight from.

So the corrected sixth theorem is a theorem about background-relative repugnance. What ED, GNEP, NE and WNS jointly force is VRA^* : for each quantity of misery, some background against which that misery plus enough marginal lives beats excellent lives. On the capacity view those backgrounds are the ones that have redeemed their misery, and the conclusion is the Repugnant Conclusion for such backgrounds; and the four conditions do not force VRA , as the capacity view shows. Anyone who wants more must add WQA^* , and the only thing WQA^* adds to WQA is the stipulation that the uncompensable quantity of misery be the same for every background: by Proposition 2 it has no work to do against Archimedean views, and its content is the exclusion of views like the capacity view.

Two remarks on scope. The result does not depend on denying Finite Fine-Grainedness. The known escape, Carlson's non-Archimedean totalism (2022, sec. 5), Thomas's total lexic utilitarianism (2018, sec. 1), Thornley's Lexical Totalism, gives welfare levels a lexical structure so that no finite sequence of slight steps leads from very high to very low; Thornley says "the theorem is only true given Finite Fine-Grainedness" (Thornley 2021, sec. 3), and is right about the strengthened theorem. The capacity view lives on integer levels, where every level is finitely many steps from every other, and its lexical structure is in the ranking of populations, not of lives, which the four fine-grainedness-using conditions do

not reach. And the view also satisfies Dominance Addition, Non-Sadism and Mere Addition; it is pathological only where it was designed to be.

5 How the counterexample was found

The capacity view was not found by trying candidates. Arrhenius’s conditions other than the Quality conditions are *transfer conditions*: each says that a population obtained from another by a specified change is at least as good, perhaps only when the background satisfies a restriction. An impossibility proof is a chain of licensed transfers ending at a population the conclusion’s denial says is no better than the start. So look for a quantity Φ on populations that no licensed transfer lowers, and rank populations first by $\min(0, \Phi)$ and then by total welfare. If Φ is additive over union, with a weight w_ℓ per level, backgrounds drop out and each condition is a linear inequality on the weights. GNEP, lowering one life from z to $z - 1$ while n lives rise from a low level y to a high level x , requires $w_{z-1} + nw_x \geq w_z + nw_y$; NE, lowering one life from x to $x - 1$ while n rise from y to $x - 1$, requires $(n + 1)w_{x-1} \geq w_x + nw_y$; WNS, comparing D lives at a positive level y' with m at a negative level x' , requires $Dw_{y'} \geq mw_{x'}$; and ED requires the weights to be non-decreasing. One transfer lowers every such Φ : Non-Elitism at the very high level, raising non-negative lives, removes an excellent life and adds nothing negative. It is licensed only in misery-free backgrounds, so it is neutralized by capping the layer at zero, provided $w_\ell \geq 0$ for $\ell \geq 0$; that is why the layer is $\min(0, \Phi)$. And refusing a conclusion of the form “for all $N, m \cdot z + N \cdot 3 + P$ beats $n \cdot A + P$ ” requires $\Phi(m \cdot z + N \cdot 3) < 0$ for all N , hence $w_3 \leq 0$, hence with the cap $w_0 = \dots = w_3 = 0$ and $w_z < 0$.

For fixed witnesses this is a linear feasibility problem in a dozen variables, settled instantly by a solver; the script accompanies the paper. For the sixth theorem’s conditions, with every witness $\mathbf{i}$, it returns $w_\ell = \ell$ for $\ell < 0$, 0 for $0 \leq \ell < A$, and 1 for $\ell \geq A$: $\Phi = H + T^-$, the capacity view. For the fifth theorem, which replaces Non-Elitism by General Non-Elitism and Weak Non-Sadism by Dominance Addition, the system is infeasible for every small witness, because the unrestricted background lets the transfer at the very high level occur in populations with uncovered misery, forcing $w_A \leq 0$, which the other conditions propagate into a contradiction. For the first four theorems the method is uninformative, since their conclusions involve no misery and an axiology of this shape can refuse a misery-free conclusion only through $w_3 < 0$, which the cap forbids; the sixth is the only theorem whose conclusion involves misery, and in the fifth General Non-Elitism closes the gap. The method finds countermodels of one shape, and infeasibility is evidence rather than proof; but when it succeeds it delivers a candidate whose verification is routine, and when it fails it shows which condition excludes. Here it delivered the capacity view in the first run, and the verification took longer than the search. That the output was a modification of a model already in the literature is some evidence that it finds natural objects; the modification, capping the layer at zero, is the one Thomas’s model needed.

6 Machine verification

The error in the sixth theorem is a quantifier-dependence error, a witness chosen for one background used for another; it is the kind a proof assistant catches and referees do not. Social choice has its impossibility theorems verified and even discovered by machine (Nipkow 2009; Tang and Lin 2009; Geist and Endriss 2011; Brandt and Geist 2016); population axiology has one mechanized result that I can find (Parent and Benzmüller 2024), and none of Arrhenius’s theorems had been checked. I have formalized the results of this paper in Lean 4 (Moura and Ullrich 2021) using only the core library; the source accompanies the paper. The theorem `sixth_theorem_countermodel` states that for every $A \geq 4$ and $c \geq 0$ the capacity view is complete and transitive and satisfies the five conditions in their exact formulations, and violates WQA*; populations are lists of integer levels, and “ $B \subset R(1, y)$ ” is the hypothesis that every member lies between 1 and y . The theorems T1 to T5 and T6star verify Thomas’s reconstructions of all six theorems, the last being that ED, GNEP, NE and WNS entail VRA*, with populations as finitely supported count functions and conditions in Thomas’s fixed-threshold forms; `prop2` and `cor3` verify Proposition 2 and Corollary 3; and `T6star_exact` repeats the corrected theorem for Arrhenius’s exact conditions, where the thresholds are chosen inside the conditions and the low population is an arbitrary list in the range. Thomas’s statements and proofs are correct in every case. Two fidelity choices are recorded in the source, besides the side condition $y \geq 3$ that Arrhenius’s definition of a range imposes: in Egalitarian Dominance the common size is positive; and in Weak Non-Sadism, where Arrhenius writes “a number of lives,” the countermodel is shown to satisfy the condition with a positive number, the stronger claim, while the theorems assume only the weaker form. All proofs are complete, contain no `sorry`, and depend only on Lean’s three standard axioms; the development is about twelve hundred lines.

7 Conclusion

Arrhenius’s sixth impossibility theorem is false. The capacity view satisfies all five of its conditions in their exact formulations, on integer-indexed welfare levels, while accepting the Repugnant Conclusion and rejecting the Very Repugnant Conclusion. The theorem becomes true when Weak Quality Addition is strengthened so that the uncompensable quantity of misery is fixed before the background, as Thomas and Thornley propose, and the strengthened theorem is machine-checked along with Thomas’s reconstructions of the other five. But the strengthening is not free. For every axiology on which misery can be compensated by lives barely worth living, Weak Quality Addition in either form entails the Quality Addition Condition the sixth theorem was meant to weaken; for the axiologies on which it cannot, the strengthened condition rules out, by stipulation, the view that how much misery a population can absorb depends on how much excellence it contains. What the four remaining conditions establish is that for any quantity of misery there is some background against which the Repugnant Conclusion holds with that misery thrown in. On the capacity view those are the backgrounds that have already redeemed it, and the verdict is the Repugnant Conclusion over again; a critic who finds the verdict repugnant whatever the background has a quarrel with the four conditions, not with Weak Quality Addition in either form.

The error survived years of citation before Thomas found it, and a decade since, because finding a gap in a proof and finding out whether the claim is true are different tasks. Population axiology has a stock of impossibility theorems whose premises are each the product of careful philosophical judgment and whose proofs are each a page of inequalities that almost nobody re-derives. The judgment cannot be automated. The inequalities can, and should be.

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